Q1:

Interation 1

distribute([9, 6, 7], A, B).

① H=9, T=[6,7], A=[9, L1], B=L2 []

?-divisible(9,2).
False

1

② 2

?-divisible(9,3).
Ture.

③ NEXT PAGE

3

interation 2

distribute([6, 7], L1, L2).

① 4 H=7, T=[6], L1=[6|L1'], L2=L2'

?- divisible (6,2)
Ture

② H=7, T=[6], L1=[6|L1'] ?-divisible (6,3)
True

③ H=7, T=[6], L2=[6|L2'] ?-% 3
True

distribute([7]..).

solution 3

A=[9]
B=[6,7]

5

interation 3

step 2: back to 6 ≥ 3 but 6 → L1

step 3

distribute([7], L1', L2').

back

① x can't

② x can't

step 6

③ H''=7, T''=[], L1'=L1'', L2'=[7|L2''] step 1

interation 5

distribute([], L1'', L2'').

step 5 7 ④

A=[9,6]
B=[7]

solution 1

solution 2

process

CONT. Situation 3

9 ?- % 3

B = [9 | L2]

distribute([6,7], L1, L2)

① ② ③

6 → L1 = [6 | L1']

Solution 4 & 5

A = [6]

B = [9, 7]

6 → L2 = [6 | L2'']

Solution 6

A = []

B = [9, 6, 7]